EXPERIMENT NO:2

DATE:09-01-2026

EXPERIMENT TITLE:

SDLC-Based Estimation, Scheduling, and Feasibility Analysis for a Budget-Constrained Smart Parking System

AIM:

To apply Software Engineering principles such as requirement analysis, cost estimation, scheduling, and feasibility analysis to design a cost-effective and constraint-compliant Smart Parking System using existing infrastructure.

PROBLEM DESCRIPTION:

Infra Tech Park is experiencing severe parking congestion, where employees spend up to 30 minutes searching for available parking spaces. This results in productivity loss and frustration. The management requires an automated Smart Parking System that can guide drivers to available parking slots efficiently. The solution must be planned without purchasing new hardware and within a limited budget and timeline. Students must justify all planning decisions using SDLC concepts and demonstrate feasibility.

GIVEN CONDITIONS / CONSTRAINTS:

Total planning duration: 2 weeks

Budget cap: $5,000

Must use existing low-resolution security cameras

Existing barrier gates with basic open/close functionality

No new sensors (ground sensors, LiDAR, IoT devices) allowed

Only software-based enhancements permitted

Final project plan must be ready for management approval

APPROACH / METHODOLOGY:

The problem was addressed using a structured SDLC approach. First, requirements and constraints were clearly identified. Existing infrastructure capabilities were analyzed to determine technical feasibility. A conceptual system design was proposed using camera-based vehicle detection. Cost estimation was performed by allocating budget to software development, cloud vision APIs, testing, and contingency. Finally, feasibility analysis was conducted to validate technical, economic, operational, and schedule aspects of the solution.

PLANNING AND ESTIMATION:

Estimated costs were calculated as follows:

Cloud vision API (trial / limited usage): $1,200

Software analysis and development effort: $2,500

Testing and validation: $700

Buffer and contingency: $600

Total Estimated Cost: $5,000

This confirms that the proposed solution remains within the specified budget limit.

SCHEDULING / DESIGN CONSIDERATIONS:

Week 1: Requirement analysis, infrastructure study, API selection, and system design

Week 2: Cost validation, feasibility analysis, risk identification, and preparation of final project plan

The schedule ensures timely completion of the planning phase within the 2-week constraint.

FEASIBILITY ANALYSIS:

Technical Feasibility: Achieved by using existing cameras and computer vision techniques for parking slot detection.

Economic Feasibility: Satisfied as total cost does not exceed the $5,000 budget.

Operational Feasibility: The system aligns with current parking operations and requires no major changes for users.

Schedule Feasibility: All planning activities are achievable within the 2-week timeline.

RESULT:

A complete SDLC-based Smart Parking System project plan was developed successfully. The system design, cost estimation, scheduling, and feasibility analysis collectively prove that the proposed solution satisfies all given constraints.

CONCLUSION:

This experiment demonstrates that Software Engineering methodologies can be effectively applied to real-world infrastructure problems. By following SDLC principles, a feasible, economical, and practical Smart Parking System plan was created without additional hardware investment, highlighting the importance of structured planning and analysis.